Homework 1

Jaisel Singh (js6897) October 16, 2026

Question 1: A First Look at Denoising Models

(a) Posterior. Both $p(x_0)$ and $p(x_1 \mid x_0)$ are Gaussian with mean linear in x_0, so (x_0, x_1) is jointly Gaussian and the posterior is Gaussian before any algebra; the work is to find its two parameters. By Bayes' rule, keeping only the factors that involve x_0,

$$p(x_0 \mid x_1) \;\propto\; p(x_0)\, p(x_1 \mid x_0) \;\propto\; \exp\left\{-\frac{x_0^2}{2\tau_0^2} - \frac{(x_1 - \sqrt{1-\beta}\, x_0)^2}{2\beta}\right\}.$$

Expanding the square and dropping $x_1^2/(2\beta)$, which is constant in x_0, the exponent becomes

$$-\frac{x_0^2}{2\tau_0^2} - \frac{(1-\beta)\, x_0^2 - 2\sqrt{1-\beta}\, x_0 x_1}{2\beta} \;=\; -\frac{1}{2}\underbrace{\left(\frac{1}{\tau_0^2} + \frac{1-\beta}{\beta}\right)}_{\lambda} x_0^2 \;+\; \underbrace{\frac{\sqrt{1-\beta}\, x_1}{\beta}}_{h}\, x_0.$$

Completing the square, any density proportional to $\exp\{-\frac{1}{2}\lambda x_0^2 + h x_0\}$ is $\mathcal{N}(h/\lambda,\, 1/\lambda)$. Here $\lambda = \big(\beta + (1-\beta)\tau_0^2\big)/(\beta\tau_0^2)$, so

$$\frac{1}{\lambda} = \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}, \qquad \frac{h}{\lambda} = \frac{\sqrt{1-\beta}\, x_1}{\beta} \cdot \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2} = \frac{\sqrt{1-\beta}\,\tau_0^2\, x_1}{\beta + (1-\beta)\tau_0^2},$$

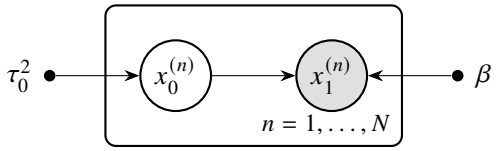
$$\boxed{x_0 \mid x_1 \;\sim\; \mathcal{N}\!\left(\frac{\sqrt{1-\beta}\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}\, x_1,\;\; \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}\right).}$$

Check. Jointly $\mathbb{E}[x_1] = 0$, $\mathrm{Var}(x_1) = (1-\beta)\tau_0^2 + \beta$ by total variance, and $\mathrm{Cov}(x_0, x_1) = \mathbb{E}[x_0\, \mathbb{E}(x_1 \mid x_0)] = \sqrt{1-\beta}\,\tau_0^2$; Gaussian conditioning, mean $\mathrm{Cov}(x_0, x_1)\, x_1/\mathrm{Var}(x_1)$ and variance $\tau_0^2 - \mathrm{Cov}(x_0, x_1)^2/\mathrm{Var}(x_1)$, reproduces both boxed expressions.

The mean is $x_1/\sqrt{1-\beta}$, the inversion of the deterministic scaling, times $(1-\beta)\tau_0^2/\big((1-\beta)\tau_0^2 + \beta\big)$, the share of $\mathrm{Var}(x_1)$ due to signal rather than injected noise; the posterior therefore shrinks that inversion toward the prior mean 0. Its variance is free of x_1: an observation relocates the posterior without changing its spread.

(b) Limits. As $\beta \to 0$ the mean tends to x_1 and the variance to 0: the step is the identity, so x_1 pins down x_0 exactly and the posterior is a point mass. As $\beta \to 1$ the mean tends to 0 and the variance to τ_0^2: x_1 is pure noise, the likelihood is independent of x_0, and the posterior falls back on the prior $\mathcal{N}(0, \tau_0^2)$. Since $d\sigma^2/d\beta = \tau_0^4/\big(\beta + (1-\beta)\tau_0^2\big)^2 > 0$, the variance climbs monotonically between these limits, so β sets how much of x_0 survives the step.

(c) Graphical model. The statement describes one (x_0, x_1) pair; the plate shows i.i.d. replication over N pairs (take $N = 1$ here). Shaded $x_1^{(n)}$ is observed, $x_0^{(n)}$ is latent, and the fixed τ_0^2, β sit outside the plate, shared across all n.



Question 2: Discrete Conjugacy

(a) Posterior. Write $\alpha_0 = \sum_{v=1}^{V} \alpha_v$ and $B(\boldsymbol{a}) = \prod_{v=1}^{V} \Gamma(a_v)/\Gamma\big(\sum_{v=1}^{V} a_v\big)$, so the prior density on the simplex $\Delta_{V-1} = \{\boldsymbol{\pi} : \pi_v \geq 0, \sum_v \pi_v = 1\}$ is $p(\boldsymbol{\pi}) = B(\boldsymbol{\alpha})^{-1} \prod_{v=1}^{V} \pi_v^{\alpha_v - 1}$; equivalently $\int_{\Delta_{V-1}} \prod_{v=1}^{V} \pi_v^{a_v - 1}\, d\boldsymbol{\pi} = B(\boldsymbol{a})$ for any $\boldsymbol{a}$ with positive entries, an identity reused in part (b). The likelihood is where the counts appear. Writing one categorical draw as $p(y_i \mid \boldsymbol{\pi}) = \pi_{y_i} = \prod_{v=1}^{V} \pi_v^{\mathbf{1}\{y_i = v\}}$ and using conditional independence given $\boldsymbol{\pi}$,

$$p(y_1, \ldots, y_n \mid \boldsymbol{\pi}) = \prod_{i=1}^{n} \prod_{v=1}^{V} \pi_v^{\mathbf{1}\{y_i = v\}} = \prod_{v=1}^{V} \pi_v^{\sum_{i=1}^{n} \mathbf{1}\{y_i = v\}} = \prod_{v=1}^{V} \pi_v^{n_v},$$

so the data enter only through the counts $(n_1, \ldots, n_V)$, a sufficient statistic, and the likelihood is a product of powers of the π_v, matching the functional form of the prior. That is exactly why the family is closed under conditioning. Multiplying,

$$p(\boldsymbol{\pi} \mid y_1, \ldots, y_n) \;\propto\; \prod_{v=1}^{V} \pi_v^{\alpha_v - 1} \prod_{v=1}^{V} \pi_v^{n_v} \;=\; \prod_{v=1}^{V} \pi_v^{(\alpha_v + n_v) - 1}.$$

The right side is the Dirichlet kernel with parameters $\alpha_v + n_v > 0$; a density is pinned down by its kernel, so the normalizer is $B(\boldsymbol{\alpha} + \boldsymbol{n})^{-1}$ and

$$\boxed{\boldsymbol{\pi} \mid y_1, \ldots, y_n \;\sim\; \text{Dirichlet}(\alpha_1 + n_1,\; \ldots,\; \alpha_V + n_V).}$$

The prior parameters behave as pseudocounts: the data simply add n_v to α_v, and the concentration grows from α_0 to $\alpha_0 + n$.

(b) Posterior predictive. Let $\boldsymbol{\alpha}' = \boldsymbol{\alpha} + \boldsymbol{n}$, so $\alpha'_0 = \sum_u \alpha'_u = \alpha_0 + n$. Given $\boldsymbol{\pi}$ the next draw is independent of the previous ones, so conditioning on $\boldsymbol{\pi}$ and marginalizing it out,

$$p(y_{n+1} = v \mid y_{1:n}) = \int_{\Delta_{V-1}} \pi_v \, p(\boldsymbol{\pi} \mid y_{1:n})\, d\boldsymbol{\pi} = \mathbb{E}[\pi_v \mid y_{1:n}],$$

the posterior mean of π_v. The extra factor of π_v raises the v-th exponent by one, replacing $\boldsymbol{\alpha}'$ with $\boldsymbol{\alpha}' + \boldsymbol{e}_v$, so that same identity applies:

$$\mathbb{E}[\pi_v \mid y_{1:n}] = \frac{1}{B(\boldsymbol{\alpha}')} \int_{\Delta_{V-1}} \prod_{u=1}^{V} \pi_u^{(\alpha'_u + \mathbf{1}\{u=v\}) - 1}\, d\boldsymbol{\pi} = \frac{B(\boldsymbol{\alpha}' + \boldsymbol{e}_v)}{B(\boldsymbol{\alpha}')} = \frac{\Gamma(\alpha'_v + 1)}{\Gamma(\alpha'_v)} \cdot \frac{\Gamma(\alpha'_0)}{\Gamma(\alpha'_0 + 1)} = \frac{\alpha'_v}{\alpha'_0},$$

where the $u \neq v$ factors cancel and $\Gamma(z+1) = z\,\Gamma(z)$ is used twice. Hence

$$\boxed{p(y_{n+1} = v \mid y_1, \ldots, y_n) = \frac{\alpha_v + n_v}{\alpha_0 + n}, \qquad \alpha_0 = \sum_{u=1}^{V} \alpha_u.}$$

These sum to one over v, since $\sum_v (\alpha_v + n_v) = \alpha_0 + n$. For $n > 0$,

$$\frac{\alpha_v + n_v}{\alpha_0 + n} = \frac{n}{\alpha_0 + n} \cdot \frac{n_v}{n} \;+\; \frac{\alpha_0}{\alpha_0 + n} \cdot \frac{\alpha_v}{\alpha_0}$$

exhibits it as a convex combination of the maximum likelihood estimate n_v/n and the prior mean α_v/α_0, with α_0 an effective prior sample size: it tends to the MLE as $n \to \infty$ but keeps positive mass on every category even when $n_v = 0$. This is additive smoothing, which becomes Laplace smoothing when $\alpha_v = 1$.

Question 3: Box's Loop

(1) Question. Among the available Boston Marathon race records, does a two-component Gaussian mixture predict late-race slowing better than a single Gaussian or a single skew-normal, and would that justify calling its components "wall" and "no wall"? These are separate questions: a useful density approximation need not identify two physiological states.

(2) Data and processing. I use the public llimllib/bostonmarathon scrape of BAA results. Its scraper retrieves individual race-result pages and exports cumulative checkpoint times in minutes, rounded to two decimals. The 2013 and 2014 files contain 16,164 and 31,984 rows; variables include bib, recorded sex, age, and times at 5, 10, 20, 21.0975, 25, 30, 35, 40 and 42.195 km. A row is a race performance, not necessarily a distinct person across years; the exported files omit the 15 km split.

I remove 105 wheelchair records, seven repeated bibs within a year, and 41 records with an available segment pace outside the screening range 2–20 min/km. Missing tokens and nonpositive times become missing; non-increasing cumulative times would also be blanked (none occur here). The loader interpolates isolated interior gaps, but the analysis requires all four defining endpoints to be observed, excluding another 304 records: 47,691 remain, 457 fewer than the raw files. A missing split's cause is unknown, and the slow-pace cutoff can remove a real stop, so these are selection decisions rather than proof of faulty timing.

For cumulative time T_d at distance d, define

$$a_i = \frac{T_{25,i} - T_{10,i}}{15}, \qquad r_i = \log\left(\frac{(T_{40,i} - T_{30,i})/10}{a_i}\right), \qquad y_i = r_i - \widehat{q}_{.25}(a_i).$$

Here a_i is baseline pace and $r_i > 0$ means the 30–40 km window was slower. Negative y_i means less slowing than the reference, not necessarily a faster late pace. The reference is the training sample's 25th percentile of r in 40 baseline-pace bins, interpolated between bin medians and held constant beyond them. Figure 1a explores the raw relationship; panels b–c show the centered distributions. This is empirical centering, *not* a terrain correction: the 30–40 km window still includes Heartbreak Hill (BAA course map). Pace reflects strategy and fatigue as well as ability, and shifting different pace strata can change the pooled shape.

(3) Scope and limitations, set before the refit. These records can assess density fit in the observed sample, but contain no direct observation of glycogen depletion, fueling, pain or perceived exhaustion. The source excludes nonfinishers and acknowledges missing finishers; the 2013 race interruption adds severe selection, so neither a population wall rate nor a lower bound on it is identified. Boston's entry rules and course further limit generalization. Nonfinisher splits and independently collected fueling, heart-rate and exhaustion records would help distinguish timing patterns from a physiological event.

(4) Model and fitting. For each recorded-sex × year group g, I fit $\mathcal{M}_0 : \; y_i \sim \mathcal{N}(\mu_g, \sigma_g^2)$, a single skew-normal $\mathcal{M}_s$, and the latent-class model

$$z_i \sim \text{Bernoulli}(\pi_g), \qquad y_i \mid z_i = k \sim \mathcal{N}(\mu_{k,g}, \sigma_{k,g}^2), \quad \mu_{0,g} < \mu_{1,g}.$$

I split records 80/20 within groups (seed 20261016; 38,152 training, 9,539 held out), estimate the reference only on training records, and fit by maximum likelihood: closed form for $\mathcal{M}_0$, six-start numerical optimization for $\mathcal{M}_s$, and six-start EM for the mixture, alternating probabilistic component assignments with weighted parameter updates. To prevent the mixture's unbounded likelihood at a collapsed component, each standard deviation is constrained to at least $\max(10^{-4}, 0.02 s_g)$, where s_g is the training standard deviation; all selected fits converge and none uses that floor.

(5) Results and criticism. For each held-out record I compute $d_i = \log \widehat{p}_2(y_i) - \log \widehat{p}_s(y_i)$; positive values favor the mixture. The table reports $\sum_i d_i$ with approximate standard error $\sqrt{\sum_g n_g s^2_{d,g}}$, respecting fixed group counts, treating records as independent and holding the trained fits fixed. These errors do not include training-set uncertainty, repeat-runner dependence or variation across races.

Reference	Mixture minus skew-normal (nats)	Standard error
25th percentile (primary)	244.9	48.1
None (raw log ratio)	108.6	42.7
10th percentile	194.6	46.4
50th percentile	280.8	47.8

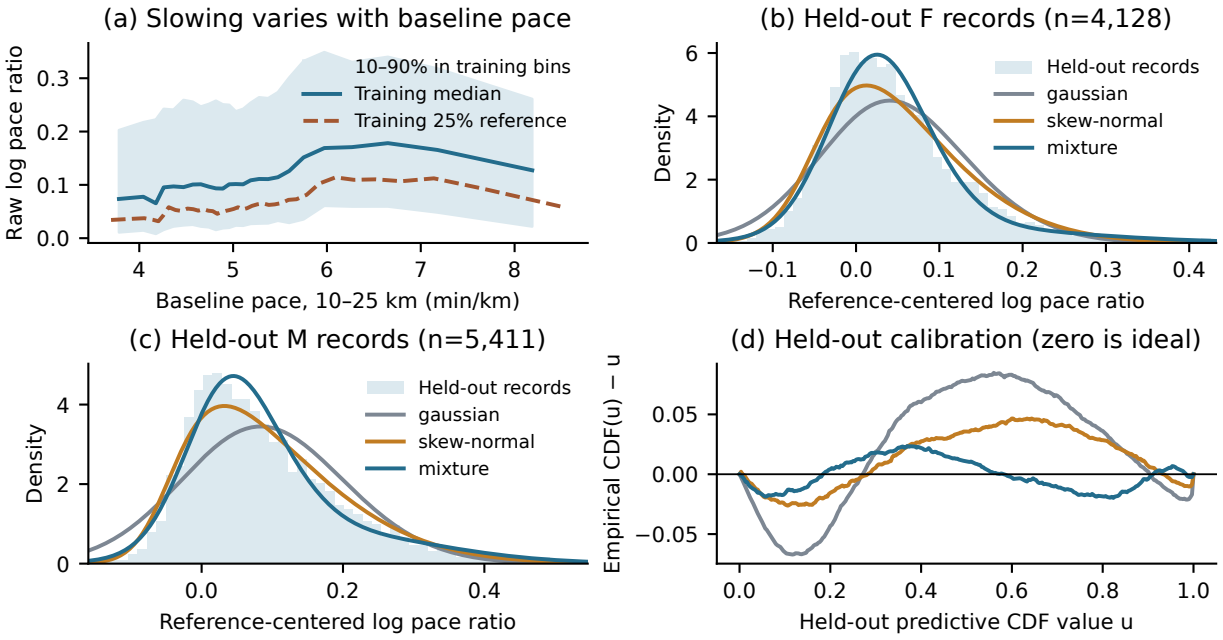


Figure 1: (a) Training quantiles; (b–c) held-out histograms and fitted densities, zoomed to the bulk; (d) predictive-CDF calibration, where zero is ideal. Scores include all tails; the mixture still shows misfit.

The primary gain is 0.0257 nats per record over the skew-normal, and 1151.5 total nats (s.e. 72.7) over the Gaussian. The advantage persists across references and in the 2014 held-out subset (179.9 nats, s.e. 42.1, using the shared training reference). Yet all 4 fitted group mixture densities have only *one* mode on a dense grid. Their higher-mean components are wider and dominate both extreme tails, including negative y, so component membership does not identify a directional slowdown. The mixture improves these density approximations; it establishes neither two biological regimes nor wall prevalence. A skew-normal is only one possible unimodal alternative.

(6) Next step. I would compare segment trajectories with independently recorded exhaustion onset in a new, uninterrupted race, including nonfinishers. The refined question is whether an abrupt, sustained slowdown coincides with that observation after accounting for course location and initial pacing. This tests what the latent label means. The current split is an internal diagnostic: an earlier draft explored these data, and the sensitivity rows reuse the test set, so neither is independent confirmation.